

Graphene oxide/polyvinylpyrrolidone composite coating on 316L SS with superior antibacterial and anti-biofouling properties

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ABSTRACT

Here, we report the fabrication of a graphene oxide (GO)/polyvinylpyrrolidone (PVP) (GP) composite coating on 316L stainless steel (SS) with superior antibacterial and anti-biofouling performance using electrophoretic deposition (EPD). At an optimal EPD condition (10 V, 60 s), PVP and GO concentrations of 0.03 and 0.1 wt%, respectively, a uniform crack free coating with superior antibacterial and anti-biofouling properties is obtained. The morphology and composition of the composite coating were characterized by field emission scanning electron microscope (FESEM), X-ray photoelectron spectroscopy (XPS), Fourier transform infrared spectroscopy (FTIR), and Raman spectroscopy. The absence of —COOH peak in GO and GP specimen found in the XPS and FTIR confirm the partial reduction of GO by oxidative decarboxylation during the EPD. The shift in the >N—C=O bond to a higher binding energy, as compared to the pure PVP, indicated the presence of intermolecular hydrogen bonding between the PVP and GO. The antibacterial activity of the coated specimens using marine bacterial isolates of Gram-negative *Pseudomonas* sp. and Gram-positive *Bacillus* sp. showed a four order reduction in bacterial density, and a significant decrease in the biofilm thickness on the GP specimens. The enhanced antibacterial activity was due to the disruption of cell membranes by sharp edges of GO surface and oxidative stress imparted due to reactive oxygen species. The enhanced surface hydration by PVP inhibited the adhesion and growth of organic molecules on GP coating. Further, the enhanced photocatalytic activity of GO due to the presence of PVP also increased the generation of H_2O_2 in GP coatings as compared to GO coating. These results show that GP composite coating is a promising candidate to tailor 316L SS surface with excellent antibacterial and anti-biofouling properties.

1. Introduction

AIISI type 316L stainless steel (SS) is a widely used structural material for cooling water services in the chemical, petrochemical, nuclear power industries, and bio-implants due to its excellent mechanical properties, corrosion resistance, ease of processability, and biocompatibility [1–4]. All-natural aqueous environments are rich sources of microorganisms such as bacteria, algae and fungi, which tend to attach and colonize on the surfaces of structural materials. The secretion of polysaccharides and other exo-polymeric substances (EPS) helps them to bind on the surfaces and these exo-polymers together with the micro-organisms form a complex microbiological matrix called biofilm [5]. The consequences of biofilm formation have been widely recognized due to its deterioration

of material properties and structural integrity. The biofilm formation and its growth (biofouling) and corrosion beneath these biofilms have been causing huge economical losses in many industries that necessitate periodic maintenance and repair of metallic structures. Therefore, it is very important not only from a research point of view, but also from an economic one, to have a deeper understanding of the mechanisms of biofilm formation, and also to develop economically viable techniques for the mitigation of microbiologically influenced corrosion (MIC).

The interaction between materials and bacteria plays an important role in biofilm formation. Studies report that the initial backbone of biofilm formation is a very thin organic film, which is known as “conditioning protein layer” which chemotactically attract bacteria, the primary colonizers of biofilm to attach [6]. According to the model

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proposed by Stanley et al. [7], the highly motile pioneer bacteria would attach to the metal surface by repeated physical adsorption and desorption process. The sticky EPS secreted by bacteria helps them to irreversibly attach to the surface. The secondary colonizers like microalgae and fungi would then attach and form a complex biofilm. All the surfaces are prone to biofilm formation depending on specific and non-specific interactions between the surface and bacteria [8]. The most effective biofilm mitigation strategy is the inhibition of biofilm formation at the beginning stage.

In this context, many strategies have been developed to overcome the issues of bacterial adhesion, colonization, and subsequent biofilm formation, which include the use of organic polymers, quaternary ammonium salts, and inorganic (copper, zinc, silver, etc.) materials and antibacterial coating [9–12]. However, leaching of metal ions and other coating components often makes the coating systems fall short of their intended use, besides causing toxicity and environmental issues [13]. In recent years, graphene derivatives such as graphene oxide (GO) and reduced graphene oxide (rGO) has gained much attention because of their ability to inhibit bacterial growth. The antibacterial activity of graphene nanomaterials depends on various factors such as size, shape, electronic properties, surface chemical properties, as well as other external parameters such as contact time, concentration, and medium [14–16]. GO based polymer composite coatings are gaining attention presently in the field of biofilm and MIC control [17–20].

Poly(N-vinyl pyrrolidone) (PVP) is a widely used polymer for coating applications in chemical, food, and biomedical industries due to its good chemical stability, lubricity, film-forming ability, complexing capability, physiological inertness, colloids characteristics, biocompatibility, anti-adhesion property, and solubility in water [21]. Studies show that PVP is a promising hydrophilic, and antibacterial surface modifier [22]. Further, studies show that PVP and GO can form a homogeneous phase due to the strong hydrogen bonding between them [21,22]. Xiong et al. [23] reported that the hydroxyl end groups of PVP can serve as a very mild reducing agent and also as a stabilizing agent. Ma et al. [24] fabricated PVP/GO thin films on the quartz crystal microbalance electrode for NH_3 detection at room temperature. Zhang et al. [25] developed a PVP modified GO composite to improve the mechanical, thermal conductivity and solvent resistance properties of natural rubber. For the fabrication of graphene based coating on various substrates, techniques such as Langmuir–Blodgett method [26], chemical vapor deposition (CVD) [27], and electrophoretic deposition (EPD) [20,28] have been attempted. Among these techniques, EPD is proven to be one of the most facile and cost-effective technique. Table 1 shows the various composite coatings prepared on SS specimens using different methods for the medical application which are tested for pathogens such as *Escherichia*

coli (*E. coli*) and *Staphylococcus aureus* (*S. aureus*). However, there are no studies on the development of antibacterial and anti-biofouling coating on SS for seawater applications.

Though there are some studies on antibacterial activities of GO based coating, systematic studies on composite materials based on GO and hydrophilic polymers such as PVP and their interaction with bacterial cells are scarce. “The novelty of this work is realization of superior antibacterial and anti-biofouling performance in the nanocomposite coating by exploiting the biocidal ability of GO and the anti-biofouling property of PVP. From the experimental results, the mechanism of the antibacterial and anti-biofouling property of the new coating is elucidated for the first time. Therefore, the present study is the first of its kind and is expected to be highly appropriate to mitigate biofilm formation on structural materials that are in contact with water bodies.”

2. Experimental methods

2.1. Materials and reagents

The chemical composition of 316L SS, the metallic substrate chosen for this study is shown in Table 2. Specimens with dimensions of $35 \times 25 \times 3$ mm were ground with SiC emery sheets up to 1000 grit finish to obtain a smooth and shiny surface. After grinding, specimens were cleaned ultrasonically with distilled water and acetone, successively for 10 min, air-dried, and kept in desiccators until further studies. To prepare GO, graphite flakes with a purity of 90% (NICE chemicals, India) were used. Polyvinylpyrrolidone (PVP) (MW 40,000) (SRL, India) was used in this study without further chemical treatment.

2.2. GO/PVP (GP) composite deposition by electrophoretic deposition (EPD)

GO was prepared by a modified Hummers' method through chemical oxidation of graphite [41,42]. The XRD and SEM analysis of prepared GO flakes are discussed in Figs. S1 and S2 (Supplementary Information). The EPD parameters were varied as per Table S1 (Supplementary Information) by varying the voltage, deposition time, and PVP contents. The GO concentration was kept constant as 0.1 wt% as reported earlier [42]. An aqueous GO/PVP (GP) suspension was prepared and the suspension was stirred for 2 h and ultrasonicated for another 1 h to maximize the homogeneity. The stability of GO and GP suspensions were evaluated by zeta potential measurement using Malvern's Zeta Nanosizer (ZEN3600). Anodic EPD employed a two-electrode cell arrangement; including 316L SS specimens as the working electrode and platinum foil as the counter electrode placed parallel to each other at a

Table 1

Summary of various coatings on SS specimens using different processing methods for the biomedical applications.

Coating material	Coating method	Application	Bacteria species	Ref.
Chitosan and PEEK with HAP	EPD	Medical and environmental fields	<i>E. coli</i> and <i>S. aureus</i>	[29]
Polydopamine/Ag/perfluorodecanethiol	Immersion	Medical devices	<i>E. coli</i> and <i>S. aureus</i>	[30]
Zinc/nitobium oxide	Dip coating	Biomedical applications	<i>E. coli</i> and <i>S. aureus</i>	[31]
poly(etheretherketone)/strontium, cerium/hydroxyapatite composite	Electrodeposition	Orthopedic application	<i>E. coli</i> and <i>S. aureus</i>	[32]
Copper–silver alloy	Electroplating	Medical devices	<i>E. coli</i> and <i>S. aureus</i>	[33]
TiO ₂ -polytetrafluorethylene nanocomposite	Dip coating	Metallic implants	<i>E. coli</i> and <i>S. aureus</i>	[34]
Embedding CuO Nanoparticles in PDMS-SiO ₂	Dip coating	Medical implants	<i>E. coli</i> and <i>S. aureus</i>	[35]
Silver-functionalized methyl-silica hybrid materials	Dip coating	Surgical devices	<i>E. coli</i>	[36]
Chitosan-copper nanoparticle	EPD	Biomaterials applications	<i>E. coli</i> and <i>S. aureus</i>	[37]
Silver nanoparticle/polytetrafluorethylene	Dip coating	Metallic implants	<i>E. coli</i>	[38]
Chlorhexidine-polydopamine	Immersion	Medical devices	<i>E. coli</i> and <i>S. aureus</i>	[39]
Chitosan/gelatin/silica-gentamicin	EPD	Medical implants	<i>E. coli</i> and <i>S. aureus</i>	[40]
GO/polyvinylpyrrolidone composite	EPD	Seawater application	<i>Pseudomonas</i> sp. <i>Bacillus</i> sp.	Present study

Table 2

Chemical composition of 316L stainless steel.

Elements	Cr	Ni	Mo	Mn	Si	P	C	S	Fe
% composition	16–18.5	10–14	2–3	<2	<1	<0.045	<0.03	<0.03	Balance

distance of 1 cm. Initially, deposition of the composite coatings was carried out by varying different voltage from 3 V to 10 V for 60 s at room temperature. Then composite coatings were prepared by varying time from 10 s to 120 s at 10 V. Further, the composite coatings were fabricated by varying PVP concentration from 0.005 to 0.04 wt%. The obtained composite coating was allowed to dry for 12 h at 60 °C in a hot air drying oven and designated as GP. For comparison, GO alone was coated on 316L SS at 10 V and 60 s without PVP and designated as GO.

2.3. Coating characterization

The chemical composition of the GO and GP specimens were analyzed using XPS, FTIR, and Raman spectroscopy. X-ray photoelectron spectroscopy system (SPECS, Germany) equipped with a mono-

$$\text{TVC} \left(\frac{\text{CFU}}{\text{cm}^2} \right) = \frac{\text{count of bacterial colonies} \times \text{dilution} \times \text{volume plated} \times \text{biofilm volume}}{\text{area of the specimen}} \quad (1)$$

chromatic Al K α source (1486.7 eV) was used. The survey spectra and high resolution spectra of each elemental peaks were acquired for as prepared GO and GP specimens and fitted with Gaussian–Lorentzian line shapes using CASA XPS software. Fourier transform infrared (FTIR) spectra for as prepared GO and GP specimens were recorded in the range of 500–4000 cm⁻¹ using an infrared spectrometer (Bruker, Germany) to identify different functional groups. Raman spectra were acquired for GO and GP specimens using the Multiview 2000 instrument (Nanonics, Israel) in the range of 800–2500 cm⁻¹ to investigate the chemical bonding. The morphology of coating surfaces was examined by field emission scanning electron microscopy (FESEM) (Zeiss, Germany). Water contact angle (WCA) was measured at room temperature using an OCA 15EC device (Data Physics Instruments, Germany). Water drops of 10 μ L were gently dispensed using Hamilton 500 μ L gas-tight syringe. WCA values were estimated by Young–Laplace fitting, using SCA 20U software. The measurements were carried out at 5 different places of the specimens and the mean values with standard deviation were reported. The thickness of the GO and GP coatings was estimated using a surface profilometer (M/s. Dektak XT, Bruker).

2.4. Preparation of microbial culture media

To test the antibacterial property of coated specimens, the specimens were exposed to Gram-negative and Gram-positive bacteria cultures. Major biofilm formers such as Gram-negative *Pseudomonas* sp., and Gram-positive *Bacillus* sp., were used in this present study [43]. The isolated bacterial colonies were separately inoculated into the sterile 100% nutrient broth (NB) (Peptic digest animal tissue 5 g/L, NaCl 5 g/L, beef extract 1.5 g/L, distilled water 1 L, and pH 7.4 $\pm$ 0.2) and incubated at 37 °C ($\pm$ 0.2 °C) for 24 h. After 24 h, the visible turbidity of the broth indicated the growth of the bacterial species. Then, 1 mL of the revived 24 h culture in 100% NB is transferred to a sterile 10% NB and incubated for 12 h at 37 °C ($\pm$ 0.2 °C). This diluted NB medium with $\sim 10^7$ CFU/mL of cells simulating natural environments was used as the exposure medium for biofilm formation on uncoated and coated SS specimens in the visible light.

2.5. Post-exposure analysis

2.5.1. Bacterial density analysis

The total bacterial density was determined by the total viable count (TVC) technique from the biofilms formed on uncoated and coated SS specimens exposed to bacterial culture media for 6 h in visible light. The exposed specimens with biofilm were suspended in sterile phosphate buffer solution (PBS) (KH₂PO₄ 0.0425 g/L and MgCl₂ 0.19 g/L) and sonicated for 5 min. The biofilm containing PBS solution was serially diluted and plated using nutrient agar (NA) as the bacterial growth medium as per the pour plate method [44]. The plates were incubated for 24 h at 37 °C ($\pm$ 0.2 °C) and as per APHA standard [45], the bacterial density was determined. The mean TVC in each biofilm was calculated considering the area of specimen and denoted as CFU/cm².

where dilution = 10⁻¹, 10⁻³, 10⁻⁵ and so on; volume plated = 10⁻¹ mL; biofilm volume (the volume into which the biofilm is ultrasonically removed) = 10 mL; and area of the specimens exposed to bacterial culture is 7.4 cm².

2.5.2. Microscopic examination

Coated and uncoated SS specimens withdrawn from the bacterial culture after 6 h were rinsed with sterile PBS to remove the loosely adhered cells. The specimens with biofilm after air drying were stained with acridine orange (AO) (0.1 wt%) for 30 min and rinsed with sterile PBS and again air-dried in dark condition and observed using Epi-fluorescence microscope (Nikon Eclipse E600, Japan). This cationic dye differentially stains single-stranded RNA and double-stranded DNA, fluorescing orange-red and green, respectively [46,47] when observed under a microscope (excitation filter BP490; Barrier filter O 515).

Confocal laser scanning microscope (CLSM) is a non-destructive and versatile technique in the characterization of biofilm architecture [48,49]. The specimens exposed in bacterial culture media for 5 d were stained targeting the total cells and their major components: proteins, amino acids, lipids, α -polysaccharides, β -polysaccharides, etc. The staining protocol is given in Table S2 (Supplementary Information) and it is followed as per the serial no [49]. After washing off the excess stains, these specimens were imaged using LSM880 Confocal microscope (Carl Zeiss, Germany). 3D and orthographic views of biofilms were reconstructed to analyze the distribution of biofilm components and thickness over the specimen surface using LSM Image Browser (Carl Zeiss Microscopy GmbH, Munich, Germany) [49]. Image analyzer J V1.45 and COMSTAT2 software were used to quantify the micro-scale biofilm structural parameters including biofilm area, biomass, average diffusion distance, and average thickness distribution [50].

2.5.3. Live/dead assay

To investigate the distribution of live and dead bacterial cells, SYTO9 and propidium iodide (PI) (BacLight™, Thermo Fischer Scientific, India) stains were used. Green fluorescing SYTO9 can enter all cells and is used for assessing total cell counts, whereas red fluorescing PI enters only cells with damaged cytoplasmic membranes. The emission properties of

these stains bound to DNA change due to the displacement of one stain by the other and quenching by fluorescence resonance energy transfer [51]. This stain often enables to differentiate only between live and dead cells. The SS specimens with 6 h old biofilms were stained with SYTO9/PI for 15 min and imaged using a fluorescence microscope and fluorescence intensities were quantified using Image-Pro Ver. 6.3 (Media Cybernetics, USA).

2.5.4. Bradford protein assay

Bovine serum albumin (BSA) standard solution was prepared at 0.1 wt% concentration and serially diluted using PBS. 4 μ L of each dilution is dispensed into the wells of the 96-well microtiter white plate. 200 μ L of pre-warmed Bradford dye reagent is added to each well and incubated for 5 min at room temperature (25 °C). A multimode reader (Glomax GM 3000, M/s. Promega Co., San Luis Obispo, CA) was used to measure the absorbance at 595 nm and a standard curve was generated. The specimens (uncoated, GO, and GP) were exposed to continuous culture for 15 d to get an adequate amount of protein from the biofilm adhered to the surface of specimens. The biofilm was collected from SS specimens in sterile PBS solution by mild sonication. The protein content from the biofilm was determined following the same protocol as described for the standard solution and the concentrations were estimated from the standard curve [52].

2.5.5. Evaluation of photocatalytic activity and bandgap

The photocatalytic activity of GO and GP specimens were assessed by evaluating the degradation of methylene blue (MB) solutions using UV–Vis spectrophotometer (Shimadzu, Japan) [53]. The uncoated and coated specimens were immersed in 20 mL of 10 mg/L MB solution and exposed under visible light for 6 h. Thereafter these specimens were removed periodically every 2 h and the absorbance of the MB solution at 665 nm was measured. For comparison, only MB was also exposed under visible light for 6 h. The bandgap of GO and GP were calculated with the aid of Tauc method from UV–Vis absorption spectra [54]. Here wavelengths of 200–700 nm were used.

2.5.6. Reactive oxygen species (ROS)-H₂O₂ assay

ROS production by bacterial cells (*Pseudomonas* sp.) exposed to GO and GP suspension was evaluated by the H₂O₂-luminol assay. Initially, the potential interference of NB (negative control) and 250 μ M of H₂O₂ solution (positive control) with ROS-Glo detection solution was evaluated in an acellular system by incubating each with 20 μ L of H₂O₂ substrate for 6 h in visible light. Similarly, 10 μ L of GO and GP suspension (0.1 wt%) in bacterial culture along with H₂O₂ substrate was incubated in 96 well microtiter white plates in visible light. Luminescence was recorded 20 minutes after the addition of 100 μ L ROS-Glo detection solution to the systems, by using a Multimode reader (Glomax GM 3000, Promega, San Luis Obispo, CA). The results are expressed as relative luminescence units (RLU), where the H₂O₂ production is proportional to the luminescence values as compared to the negative control and positive control [55]. All the experiments are repeated thrice to get reproducible results.

3. Results and discussion

3.1. Characterization of GO

XRD technique is very sensitive to interplanar crystal structure as well as the distortion occurring in layer structures. XRD pattern (Fig. S1, Supplementary information) of natural graphite showed one intense and sharp peak at 2θ value of 30.65° corresponding to the (002) plane and a low intense peak was observed at 51.5° corresponding to the (004) plane. For GO flakes a broad peak was observed at 2θ value of 10.87° corresponding to the (002) reflection, indicating the interplanar distance between the GO layers. Another peak at 2θ value of 48.97° in GO flakes corresponding to the plane (100) indicate the short range order of

stacked GO layers [56–58]. Applying the Bragg's Law to the reflection (002), the interplanar distance (d_{002}) was evaluated to be 0.34 nm and 0.95 nm for natural graphite and GO flakes, respectively [42].

Using Debye Scherrer's equation (reflection (002) and structural factor value as 0.9), the average crystal stack height (L_c) of natural graphite and GO were calculated to be 15.2 nm and 5.5 nm, respectively. The decrease in average crystal stack height indicated the successful chemical exfoliation during the oxidation of natural graphite to GO flakes. The number of layers (N_c) in the stacking structure is calculated using the equation,

$$N_c = \frac{L_c}{d_{002}} \quad (2)$$

It is found that number of stacking layer for graphite and GO flakes were found to be 44.9 and 5.8, for natural graphite and GO, respectively.

The Debye Scherrer equation was also applied to the two dimensional reflection (100) of GO flakes using the value of Warren constant as 1.84 [59] and the average in-plane crystallite size of stacking layers (L_a) was estimated to be 24.97 nm. Similar interplanar spacing of crystal stacking layers, average in-plane crystallite size of stacking layers, average crystal stack height, and number of layers present in GO flakes prepared by modified Hummers' method are reported in literature [56–58].

3.2. Optimization and formation mechanism of GP film by EPD

The pH and zeta potential of GO suspension were found out to be $\sim$ 3.4 (acidic) and -44 mV, respectively. The negative zeta potential value of GO suspension shows that the surface oxygenated functional groups of GO sheets were deprotonated. The pH and zeta potential of GP suspension was found to be $\sim$ 3.4 (acidic) and -38 mV, respectively. At acidic pH, PVP is positively charged because of the protonation of pyrrolidone nitrogen atom [60]. This neutralizes the negative charge of the carboxylate ion produced upon deprotonation of GO and thereby reducing the zeta potential to -38 mV in the GP suspension. Basically, suspensions with zeta potential values higher than ± 30 mV are considered stable [61]. In the present case, the zeta potentials of all suspensions were above ± 30 mV, indicating its good stability.

In EPD processes, the quality and quantity of deposits are controlled by varying the suspension characteristics and electric field [62]. The coating parameters like voltage and deposition time were optimized to obtain uniform coverage on the surfaces. The coating fabricated at 3 V and 5 V for 60 s were not uniform. This was because at lower voltages, the lower deposition of GO lead to poor packing between particles and non-uniform coating. With increasing voltage to 10 V, the deposition rate of composite particles was enhanced that enables better packing between the depositing GO particles, and thereby effective void filling occurred. Su et al. [63] have also reported a reduced porosity of EPD deposits when the voltage was increased from 5 to 20 V. Though the coating was not uniform throughout the deposition times of 10 s and 30 s, a uniform coating was achieved at deposition time of 60 s. Further increase in the deposition time to 120 s, resulted in the formation of thick coatings on the specimens with some delamination. These results show that the deposition time during EPD has an important role in the quality and quantity of films formed. Shorter deposition time leads to poorer GO sheet stacking, leading to a non-uniform coating, while longer deposition time can lead to increased packing with insufficient attachment to the substrate, resulting in delamination [64]. Hence an optimal deposition time leads to greater packing of GO films [64].

To optimize the PVP content in EPD suspensions, GP composite with different concentrations of PVP was deposited at a constant voltage of 10 V and deposition time of 60 s. The photographs of different PVP loaded composite coatings are shown in Fig. S3, of Supplementary information. Wajid et al. [65] reported that PVP can be used to prevent GO aggregation in aqueous dispersions and to produce a uniform coating. In our study, we observe that even at a lower concentration of PVP (0.005

wt%), a uniform composite coating can be obtained without any aggregation of GO. When the PVP loading is increased to 0.03 wt%, the coating was uniform throughout. However, at a PVP loading of 0.04 wt %, coating showed irregular packing and flaking. Amrollahi et al. [66] also indicated that increasing PVP content leads to a lowering of zeta potential of the suspension, due to reduced inter-particles repulsion that results in flocculation of particles and flaking. Therefore, EPD parameters (10 V, 60 s) and 0.03 wt% of PVP were taken as the optimum values for developing a uniform coating in the present study.

3.3. Compositional analysis

Fig. 1a–f depicts the XPS spectra of the GO and GP specimens. The photoelectron peaks corresponding to C and O elements are observed in the survey spectrum of GO specimen (Fig. 1a), whereas C, O, and N elements are observed in the survey spectrum of GP specimen (Fig. 1b). The high-resolution spectra acquired for each elemental peak are deconvoluted using CASA software to obtain the detailed surface composition and chemical state. The deconvoluted C 1s core level peak

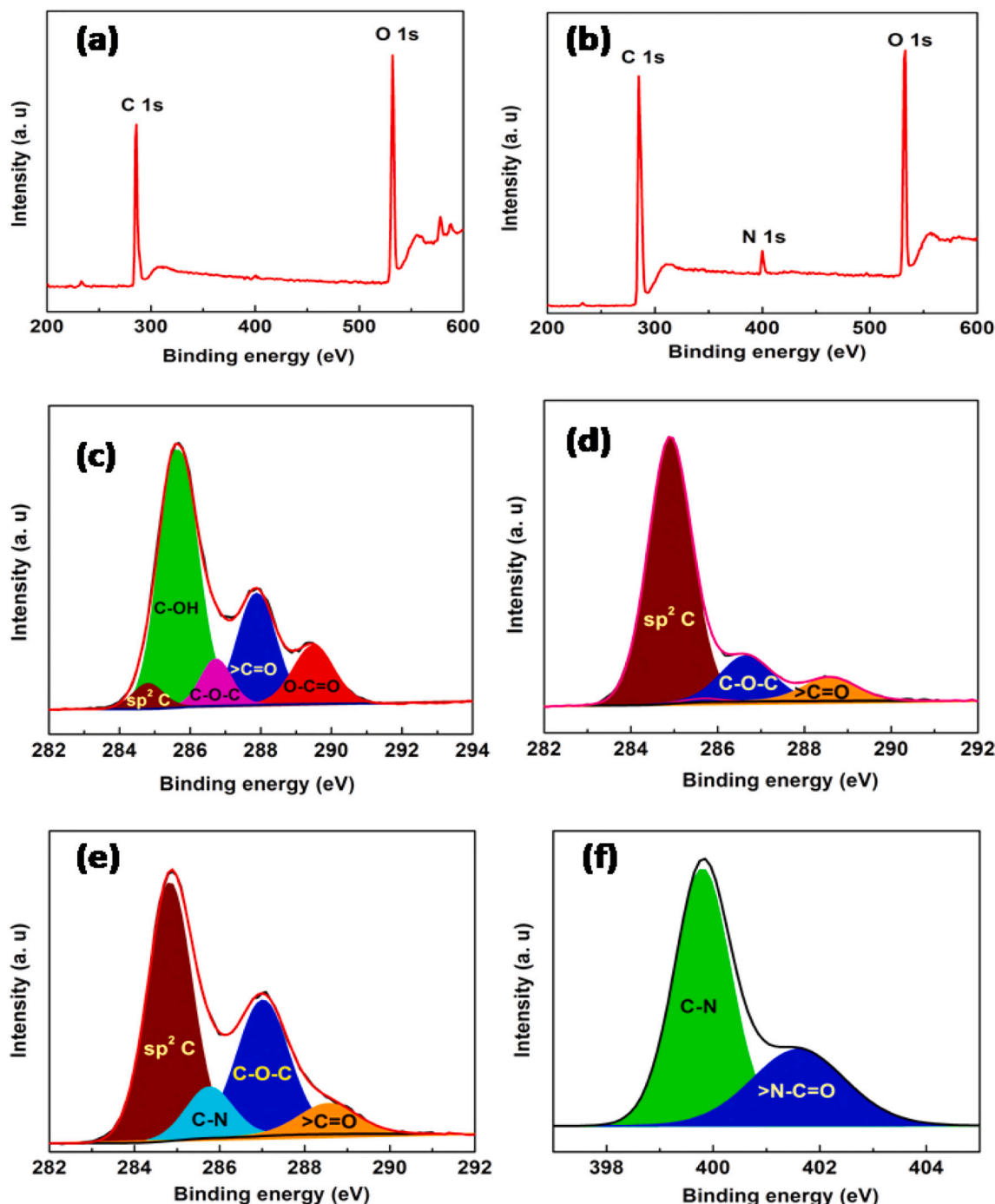


Fig. 1. XPS survey spectrum of (a) GO and (b) GP specimens, high-resolution spectrum of (c) C 1s of synthesized GO flakes, (d) C 1s of GO coating, (e) C 1s of GP coating, and (f) N 1s core-level spectrum of GP coating.

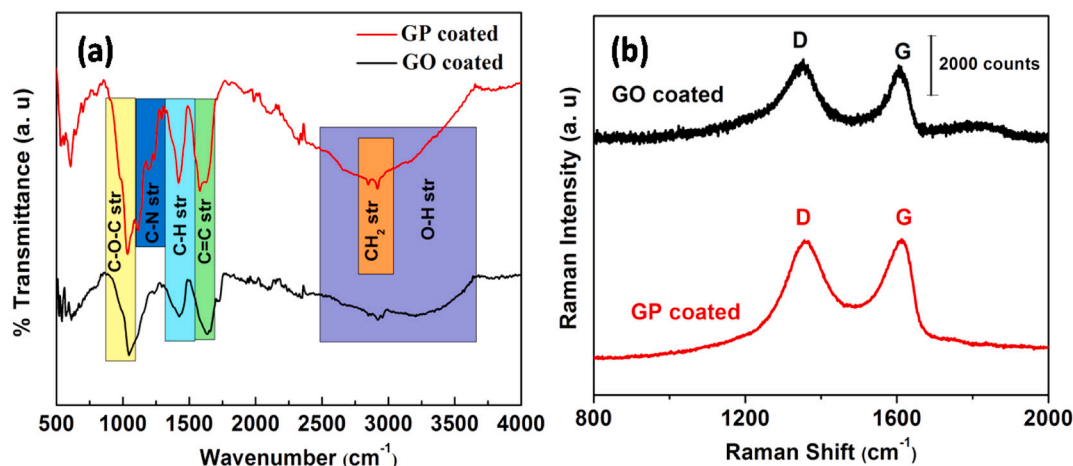


Fig. 2. (a) FTIR spectra and (b) Raman spectra of GO and GP specimens.

of synthesized GO flakes (Fig. 1c) consists of C=C, C—OH, C—O—C, >C=O, and —COOH bonds corresponding to 284.8 eV, 285.7 eV, 286.8 eV, 287.9 eV, and 289.5 eV, respectively. The deconvoluted C 1s core level peak of GO specimen (Fig. 1d) indicates the presence of C=C (284.9 eV), C—O—C (286.6 eV), and >C=O (288.5 eV) bonds. The deconvoluted C 1s core level peak of GP specimen (Fig. 1e) indicates the presence of C—N (285.8 eV) along with other bonds corresponding to the GO specimen. The intensity of C=C peaks in GO and GP specimens is found to increase, in comparison to the synthesized GO flakes, indicating the partial restoration of C=C skeleton upon reduction of oxygen functionalities during the EPD process. The absence of —COOH peaks in both the GO and GP specimens confirmed the oxidative decarboxylation during the EPD process. Sung et al. [67] also reported the reduction of oxygen functionalities during the EPD process attributed to the loss of carboxylates in Kolbe-like decarboxylation reaction. The N 1s core level spectrum was deconvoluted into two doublets (Fig. 1f). The peaks at 399.8 eV and 401.6 eV can be assigned to C—N and >N—C=O bonds, respectively [25,68]. It is observed that the >N—C=O bond shifts to higher binding energy as compared to the pure PVP (398.4 eV) [25,68], indicating the presence of intermolecular hydrogen bonding between PVP and GO.

The FTIR spectrum of GO specimen (Fig. 2a) showed peaks at 1052 cm⁻¹, 1425 cm⁻¹, and 1637 cm⁻¹ corresponding to the C—O—C, C—H, and C=C bond stretching, respectively. The FTIR spectrum of GP specimen (Fig. 2a) showed peaks at 1115 cm⁻¹ and 1289 cm⁻¹, corresponding to C—N bond stretching and deformation mode, respectively along with other peaks for GO specimen [25]. The absence of —COOH bond also noticed in both the GO and GP specimens confirmed the

oxidative decarboxylation during the EPD process. The spectrum showed a peak at 2919 cm⁻¹ and 2842 cm⁻¹ corresponding to C—H symmetric vibrations and asymmetric stretching vibrations present in the PVP chain. The presence of a broad peak at 2500–3500 cm⁻¹ depicted the presence of a large amount of physisorbed water molecules and intermolecular hydrogen bonds between PVP and GO [25].

The Raman spectra of the GO and GP specimens are shown in Fig. 2b. The GO specimen showed characteristic peaks at 1349.3 cm⁻¹ (D band) and 1606.6 cm⁻¹ (G band), associated with the disorder in GO structure and the in-plane vibration of sp² carbon atoms, respectively [69]. The wavelength shift of 6–10 cm⁻¹ is observed in the G and D bands for GP specimen. The shift is due to the high distortion of GO structure on grafting with PVP [25], the difference in bond distance of C—C and C—N bonds in GP specimens [70], and due to the presence of hydrogen bonding between GO and PVP molecule [25]. The calculated I_D/I_G value for GO and GP specimens are found to be ~1, which is slightly higher than the synthesized GO flakes (0.94) [42] that confirm the reduction of oxygen functionalities in GO during the EPD process with the restoration of sp² domain and creation of low defect density regime in GO [67].

3.4. Microstructural analysis

Fig. 3 shows the FESEM images of GO and GP specimens where the GO specimen (Fig. 3a) exhibited the continuous crumpled wavy layers over a large area due to the layer-by-layer alignment of GO sheets by the EPD process. Hasan et al. [71] reported that the deposition occurs at the anode at acidic pH gives rise to intermittent wrinkles. The sheets appeared to lie flat and parallel to the anode surface. GP films (Fig. 3b)

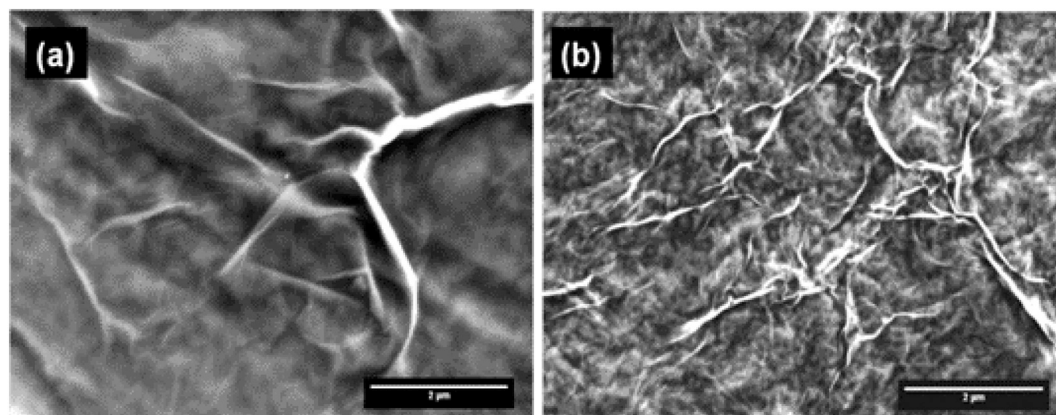


Fig. 3. FESEM images of (a) GO and (b) GP specimens.

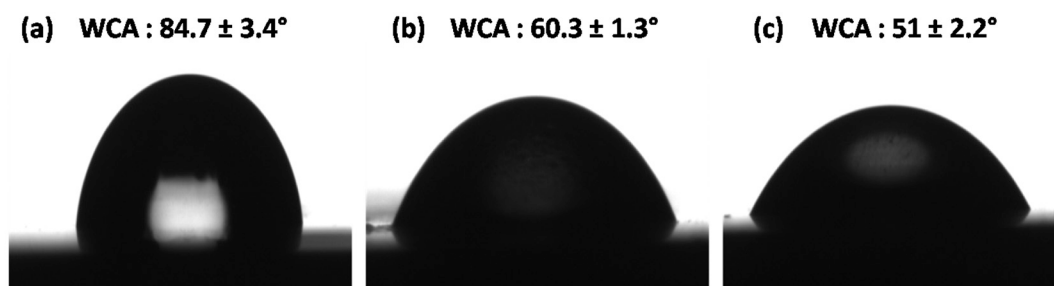


Fig. 4. Water contact angle (WCA) value of (a) uncoated, (b) GO, and (c) GP specimens.

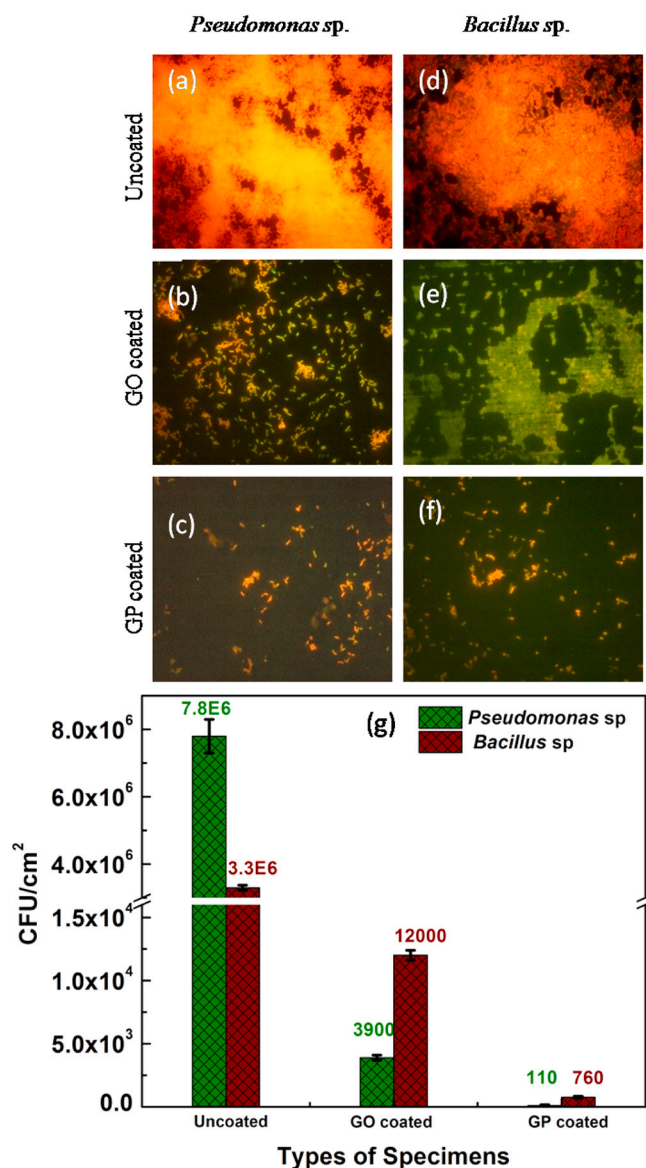


Fig. 5. Epifluorescence micrographs of the biofilms of (a–c) *Pseudomonas sp.* and (d–f) *Bacillus sp.* attached on the uncoated, GO, and GP specimens, respectively for 6 h. (g) TVC analysis on the biofilms of *Pseudomonas sp.* and *Bacillus sp.* attached on the uncoated, GO, and GP specimens for 6 h.

showed an enhanced wrinkled morphology on the addition of PVP. It is reported that PVP helps in exfoliation of GO sheets in aqueous solution [72], which results in an enhancement in layer-by-layer alignment and much irregular wrinkled microstructures [73]. The thickness of the GO

and GP coating were found to be 3.1 ± 0.6 and 3.6 ± 0.4 μm , respectively.

Fig. 4 shows the static WCA images of uncoated and coated SS specimens. The contact angle values of uncoated, GO and GP specimens were $84.7 \pm 3.4^\circ$, $60.3 \pm 1.3^\circ$, and $51 \pm 2.2^\circ$, respectively. The GO specimen exhibited a lower contact angle compared to uncoated specimens due to the presence of polar groups and the hydrophilic nature of GO film. The contact angle value was least in GP coating due to the presence of hydrophilic amino and carbonyl groups in PVP. The presence of the pyrrolidone ring in PVP behaves like a tunnel for the diffusion of water molecules through the film [74].

3.5. Antibacterial studies

3.5.1. Microscopic analysis

Epifluorescence micrographs of biofilms formed on uncoated, GO, and GP specimens by 6 h exposure in the culture of gram-negative *Pseudomonas* and gram-positive *Bacillus sp.* culture are presented in Fig. 5a–f. The uncoated surface showed maximum orange-red fluorescence indicating higher density of *Pseudomonas sp.* and *Bacillus sp.* in comparison to GO and GP specimens. The surface of the GO coating exhibited antibacterial activity with low orange fluorescence and high green fluorescence indicating the inactive cells with double-stranded DNA as compared to control surfaces [46,47]. The density of cells on GP specimens was least as shown by the least fluorescence confirming the excellent antibacterial activity.

3.5.2. Studies on bacterial density

The bacterial density estimated by total viable count method on uncoated, GO, and GP specimens in both bacterial cultures are plotted and shown in Fig. 5g. This indicates a four orders magnitude reduction of bacterial density on GP specimens followed by a three orders reduction on GO specimens as compared to the uncoated surfaces in *Pseudomonas sp.* culture. Similarly, the bacterial density of *Bacillus sp.* also showed four orders reduction on GP specimens followed by two orders reduction on GO specimens when compared to the uncoated surfaces. The TVC results corroborate the observations made in the microscopic examination after AO staining.

3.5.3. Studies on biofilm architecture

The CLSM micrographs of biofilms of bacteria formed on uncoated, GO and GP are depicted in Fig. 6. The thickness of biofilm was evaluated by orthogonal analysis (Fig. 7) and results confirmed least thickness on the surface of GP specimen when compared with the GO and uncoated surface. Using COMSTAT2 the biofilm architecture parameters like area of biofilm coverage, biomass, diffusion distance, and distribution of average thickness were estimated (Table 3). The biofilm coverage was lowest on GP specimens compared to the uncoated ones. The diffusion distances (the length of the shortest path from a voxel inside the biomass to the surface) were higher on uncoated specimens as compared to GP specimens. As depicted in Fig. 6, the average thickness of biofilm of gram-negative *Pseudomonas sp.* and gram-positive *Bacillus sp.* was only

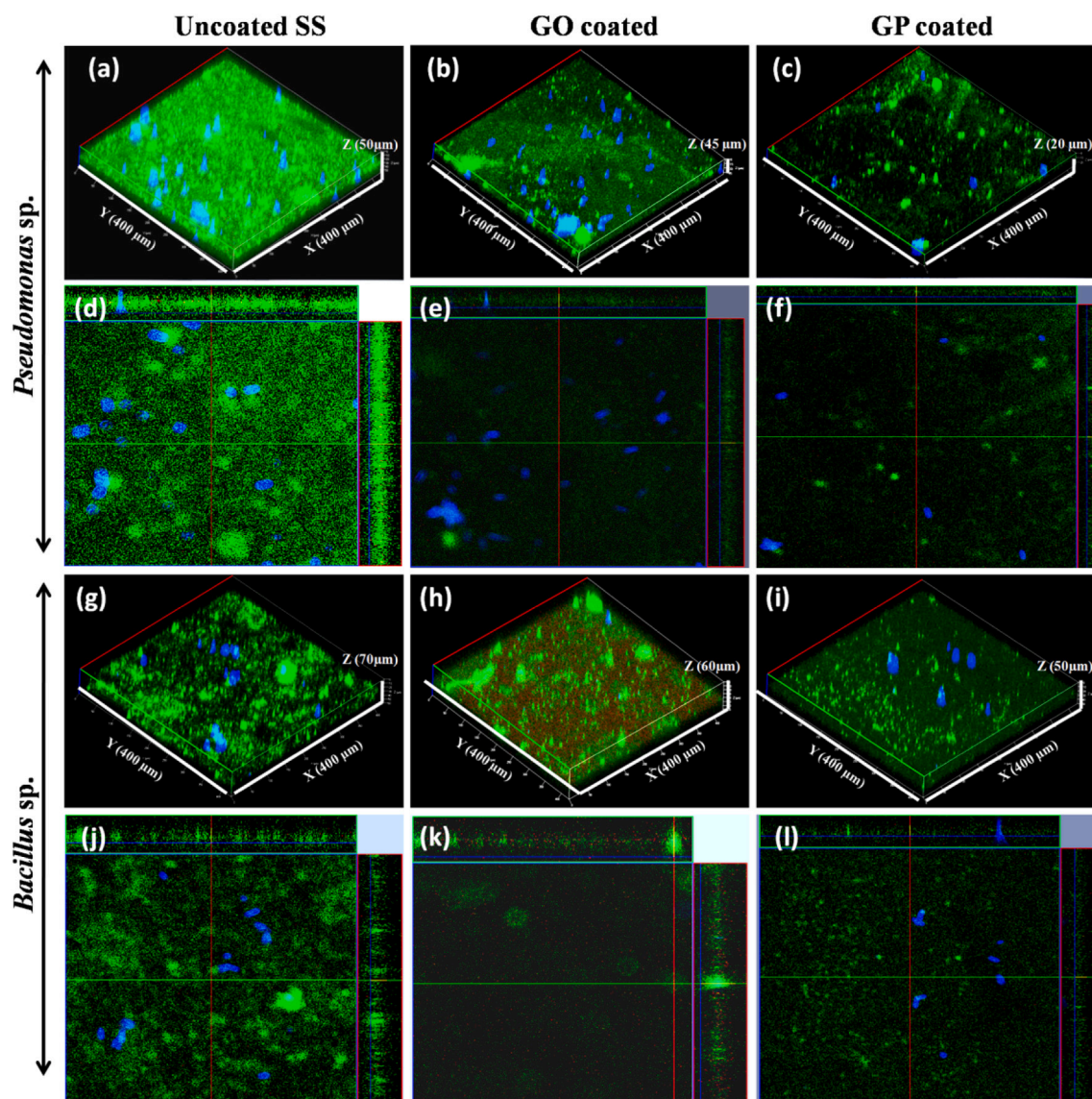


Fig. 6. Confocal [3D (a-c; g-i) and orthogonal (d-f; j-l)] micrographs of 5 d biofilms of *Pseudomonas* sp. (top panel, a-f) and *Bacillus* sp. (bottom panel, g-l) on uncoated, GO, and GP specimens.

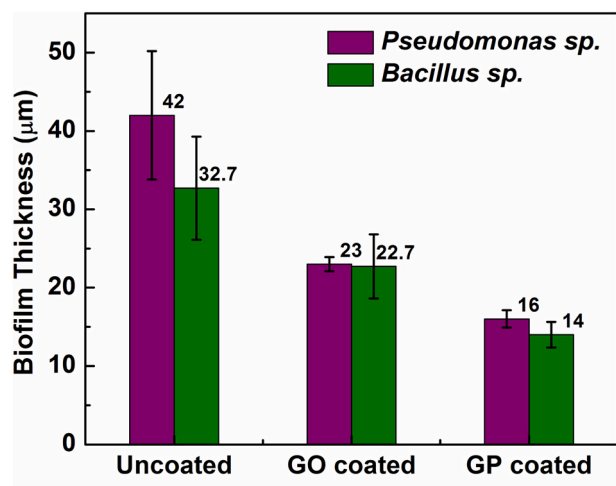


Fig. 7. Biofilm thickness obtained from the orthogonal micrograph (Fig. 6) of *Pseudomonas* sp. and *Bacillus* sp. biofilms on uncoated, GO and GP specimens.

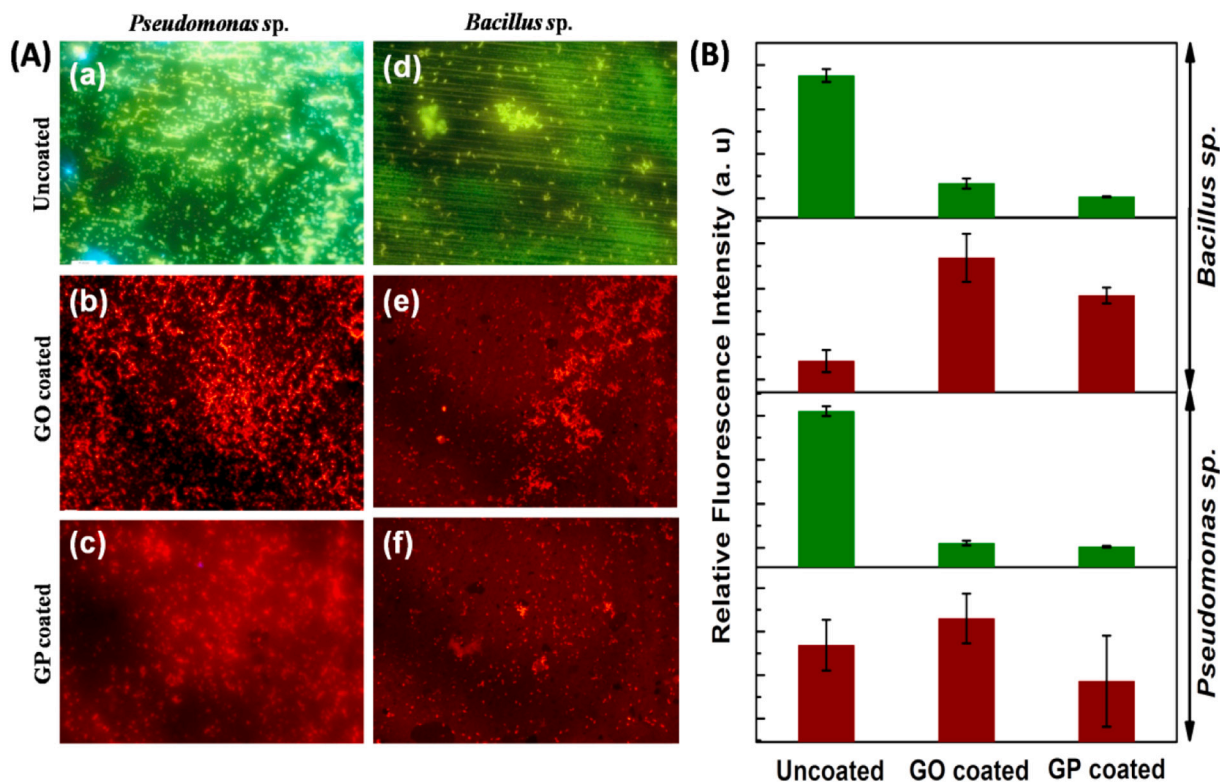
$13 \pm 2.1 \mu\text{m}$ and $16 \pm 0.3 \mu\text{m}$ on GP specimens in comparison to $34.8 \pm 6.4 \mu\text{m}$ and $41.3 \pm 8 \mu\text{m}$ on uncoated specimens, respectively. The biomass estimation clearly showed one order reduction in GO specimens and two orders reduction in GP specimen as compared to the uncoated surfaces. Thus results of CLSM analysis complemented the culture techniques and the epifluorescence microscopic analysis.

3.5.4. Live/dead assay

Epifluorescence micrographs of 6 h old biofilms of gram-negative *Pseudomonas* sp. and gram-positive *Bacillus* sp. on uncoated, GO, and GP specimens after live/dead assay along with relative fluorescence intensity are shown in Fig. 8A and B. The biofilms of both bacteria on the uncoated specimens showed intense green fluorescence confirming higher density of live bacterial cells. Higher red fluorescence by the biofilms on GO and GP specimens, indicated higher density of dead bacterial cells [75]. The relative green and red fluorescence intensity values (arbitrary units) plotted and shown in Fig. 8B further confirmed the higher inhibition of bacterial attachment and antibacterial activity on GP specimens. The least red and green fluorescence on the GP surface clearly evidenced the minimum attachment on GP surface.

Table 3COMSTAT2 analyses from confocal micrographs of uncoated, GO, and GP specimens exposed to both *Pseudomonas* sp. and *Bacillus* sp. bacterial culture for 5 d.

Bacteria culture	Specimens	Biofilm parameters				
		Area occupied (μm^2)	Biomass ($\mu\text{m}^3/\mu\text{m}^2$)	Diffusion distance (μm)	Thickness distribution	
					Average thickness of entire area (μm)	Average thickness of biomass (μm)
<i>Pseudomonas</i> sp.	Uncoated SS	1197.9 $\pm$ 12.6	0.8 $\pm$ 0.1	27.03 $\pm$ 2.2	37.9 $\pm$ 5.9	34.8 $\pm$ 6.4
	GO	68.9 $\pm$ 7.5	0.05 $\pm$ 0.03	18.19 $\pm$ 0.4	23 $\pm$ 5	23.1 $\pm$ 4.2
	GP	32.4 $\pm$ 9.4	0.003 $\pm$ 0.0014	11.25 $\pm$ 1.4	16.8 $\pm$ 1.3	13 $\pm$ 2.1
<i>Bacillus</i> sp.	Uncoated SS	180.6 $\pm$ 2.6	0.02 $\pm$ 0.004	9.9153 $\pm$ 0.2	41.7 $\pm$ 5.5	41.3 $\pm$ 8
	GO	86.9 $\pm$ 3.5	0.003 $\pm$ 0.002	8.5663 $\pm$ 0.2	20.2 $\pm$ 1.6	22.1 $\pm$ 2
	GP	27.5 $\pm$ 4.4	0.0003 $\pm$ 0.0003	2.613 $\pm$ 0.2	15 $\pm$ 0.7	16 $\pm$ 0.3

**Fig. 8.** (A) Epifluorescence micrographs of live/dead assay on (a–c) *Pseudomonas* sp. and (d–f) *Bacillus* sp. biofilm attached on the uncoated, GO, and GP specimens for 6 h. (B) Comparison fluorescence intensity of the live/dead *Pseudomonas* sp. and *Bacillus* sp. biofilm, quantified from panel (A).

3.5.5. Photocatalytic activity and ROS production

The photocatalytic reduction of MB was measured for 6 h at an interval of every 2 h with uncoated and coated SS specimens under visible light as shown in Fig. 9. Fig. 9a showed the photograph of MB solution after 6 h exposure with uncoated, GO, and GP specimens. MB solution with GP specimens showed the highest decolorization in comparison to GO and uncoated specimens. Fig. 9b represents the absorbance of MB solutions with uncoated and coated specimens at an interval of every 2 h. The MB solution (concentration of 10 mg/L) used for photocatalytic activity studies showed an initial standard absorbance value of 2.2 (a. u). On exposure to visible light, a negligible difference in absorbance value is observed over the time in MB (1.95 a. u). The absorbance values variation of MB is in the order of GP > GO > uncoated, thus indicating the highest photocatalytic activity on GP specimens.

The UV–vis spectra of GO and GP are presented in Fig. S4 (Supplementary Information). The UV–vis spectra of both GO and GP showed an intense absorption peak located at 228 nm and a weak one at 302 nm corresponding to π - π^* transition of the C=C bond and the n - π^* transition of the C=O bonds of GO, respectively. No absorption peaks of PVP is seen because of low concentration of PVP. The bandgap of both GO and

GP was measured by using the Tauc model [54] and shown in Fig. S5 (Supplementary Information). The bandgap energy was calculated as follows [54]:

$$\alpha E^{\frac{1}{n}} = A(E - E_g) \quad (3)$$

where α is absorbance, E is the photon energy, A is a constant and E_g is the optical band gap, $n = 0.5$ for direct allowed transitions or 2 for indirect allowed transitions. The E_g values of GO and GP were obtained by extrapolating the linear portion of the $\alpha E^{\frac{1}{n}}$ to E curves in Fig. S5 (Supplementary Information). The band gap for GO and GP are 2.12/3.45 eV and 2.09/3.42 eV, respectively. The reported values of bandgap of GO are 2.1/3.5 eV, which is in good agreement with our results [18,76–78].

The photocatalytic generation of hydrogen peroxide (H_2O_2) by GO and GP suspension was evaluated using ROS-Glo detection solution in a cellular system (*Pseudomonas* sp.) under visible light and results are shown in Fig. 10. The luminescence in GP suspension is higher than GO suspension, which indicates an enhancement in H_2O_2 production in the presence of PVP.

The photocatalytic generation of H_2O_2 by GO is governed by the

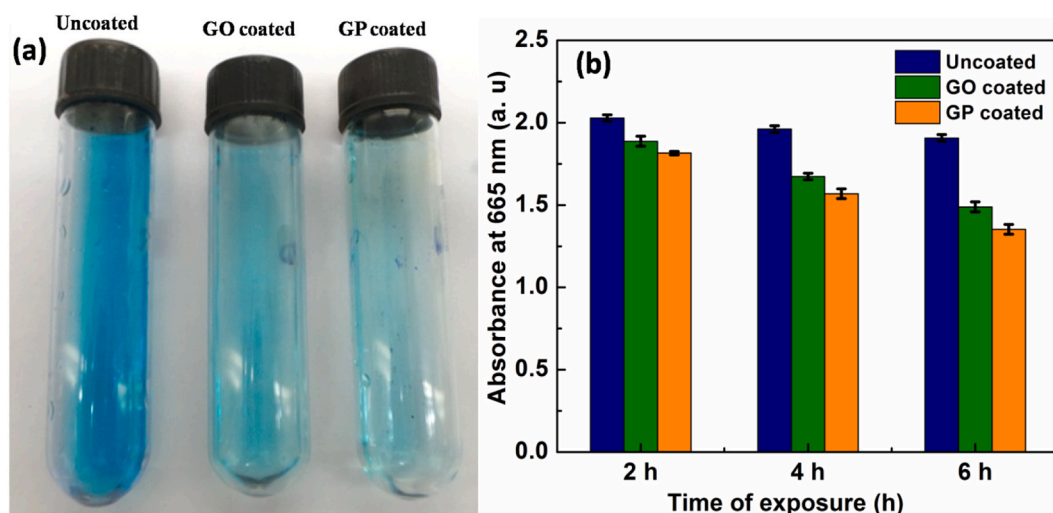


Fig. 9. (a) Photographs of MB solution after 6 h exposure at 665 nm wavelength light with uncoated, GO, and GP specimens, (b) comparison of the absorbance values of the MB solution for uncoated, GO, and GP specimens after every 2 h of exposure to visible light.

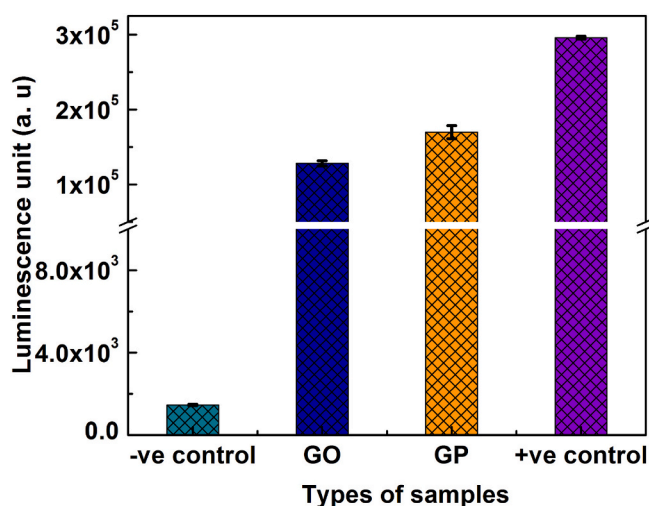


Fig. 10. Quantification of H_2O_2 production by GO and GP suspension using NB as negative control and H_2O_2 as positive control by ROS- H_2O_2 -luminol assay in *Pseudomonas* sp. The luminance values on different samples are shown.

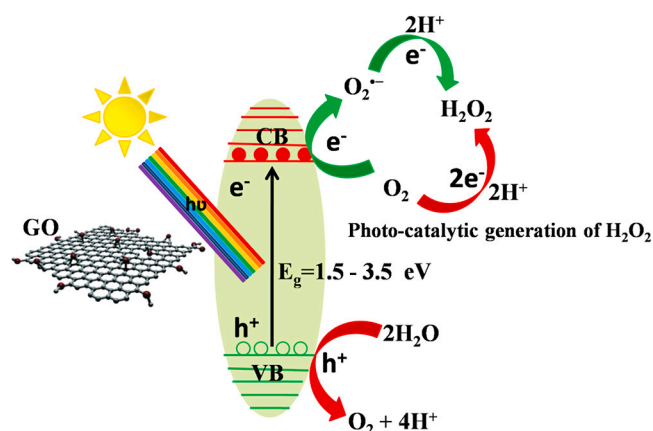


Fig. 11. Schematic representation of the photo-catalytic generation of H_2O_2 by GO.

basic principle of photocatalysis. A complete schematic illustration of a photocatalytic generation of H_2O_2 is depicted in Fig. 11. The conduction band (CB) electro-potential (E_{CB} vs. Ag/AgCl) of GO is reported to be -0.63 V at acidic medium ($\text{pH} = \sim 3$). The O_2 reduction potential ($\text{O}_2/\text{H}_2\text{O}_2$) at neutral pH is reported to be 0.08 V [79,80]. It can be seen that the CB levels remain still more negative than the reduction potential of O_2 (0.08 V) and possess the enhanced capability for O_2 reduction to H_2O_2 . The detailed discussion on H_2O_2 is given as follows. The holes (h^+) in the valance band (VB) would oxidize water molecules to form O_2 and H^+ ($2\text{H}_2\text{O} + 4h^+ \rightarrow \text{O}_2 + 4\text{H}^+$). In general, H_2O_2 can be generated through either an indirect sequential two-step single-electron reduction ($\text{O}_2 \rightarrow \text{O}_2^- \rightarrow \text{H}_2\text{O}_2$) or a direct single step two-electron reduction ($\text{O}_2 \rightarrow \text{H}_2\text{O}_2$). However, studies showed that the former pathway for photocatalytic generation of H_2O_2 is restricted in the case of GO photocatalyst [79], where they clarified that superoxide (O_2^-) was not involved in the photocatalytic generation of H_2O_2 . The overall photo-

catalytic generation of H_2O_2 ($2\text{H}_2\text{O} + \text{O}_2 \xrightarrow{h\nu} \text{H}_2\text{O}_2$) is a non-spontaneous reaction with a standard Gibbs free energy change (ΔG^0) of 117 kJ/mol [81]. The photocatalytic activity and ROS production were evident in both GO and GP coatings and the produced photoelectrons are used in the reduction of dissolved O_2 to produce H_2O_2 . Hou et al. [79] also studied the efficient generation of H_2O_2 by photocatalysis to millimolar levels under simulated sunlight and reported that dissolved O_2 played a pivotal role in the photo-production of H_2O_2 by GO. In comparison to GO specimens, GP specimens showed enhanced photocatalytic activity and thereby higher H_2O_2 production. Han et al. [82] also indicated a similar enhancement in H_2O_2 production on palladium catalysts in presence of PVP where it was reported that the PVP inhibits the O_2 dissociation and decomposition of the produced H_2O_2 resulting in high selectivity and high production rate of H_2O_2 . H_2O_2 played important role in decolorization of MB solution, where H_2O_2 was an important

Table 4

Estimated protein concentration in biofilm formed on uncoated and coated SS specimens exposed to *Pseudomonas* sp. and *Bacillus* sp. culture for 15 days.

Specimens	Protein concentration ($\mu\text{g/mL}$)	
	<i>Pseudomonas</i> sp.	<i>Bacillus</i> sp.
Bacterial culture		
Uncoated SS	918.5 ± 23	270.8 ± 23
GO	668.3 ± 38	236.8 ± 21
GP	479.9 ± 18	62.8 ± 17

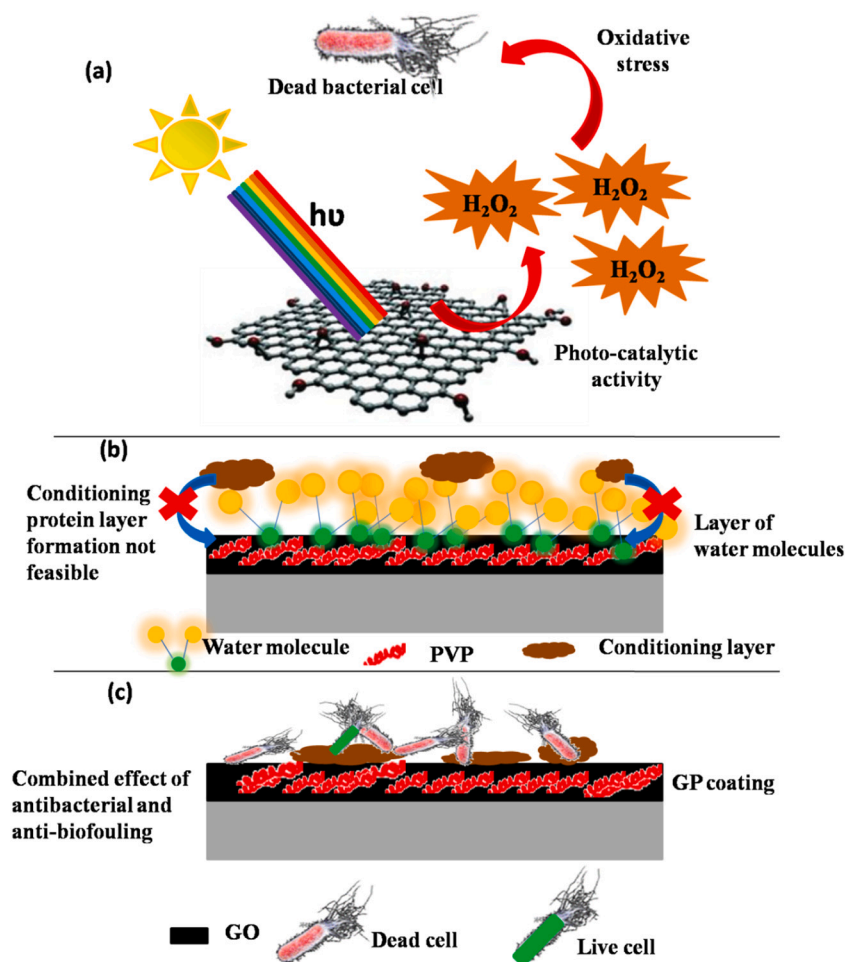


Fig. 12. Schematic representation of (a) photo-catalytic activity and ROS production by GO, (b) the anti-biofouling exhibited by PVP coating by forming a layer of water molecule, and (c) GP coating exhibited combined antibacterial and anti-biofouling property.

precursor of primary reactive species i.e. hydroxyl radical ($HO\bullet$). The $HO\bullet$ is able to initiate a series of free-radical-based reactions through the Fenton reaction, which lead to the degradation of MB [83].

3.5.6. Protein estimation

The protein content was estimated using the standard curve of BSA solution (Fig. S6, Supplementary information) and the values are listed in Table 4. It is seen that the protein concentrations of biofilm formed by both *Pseudomonas* and *Bacillus* sp. is as per the following order: GP < GO < uncoated specimens.

3.6. Mechanism of antibacterial and anti-biofouling activity

The total viable count of bacterial density, epifluorescence, and confocal micrographs, biofilm parameters, live/dead assay, and protein estimation demonstrated the superior antibacterial and anti-biofouling properties of GP specimens. The enhanced hydrophilicity and photo-catalytic activity through the generation of H_2O_2 on the GP surface were also experimentally confirmed. Based on these results the proposed mechanism for the excellent antibacterial and anti-biofouling activity by GO and PVP is schematically illustrated in Fig. 12.

The well-known modes of antibacterial activity mediated by direct contact with GO are the combined effect of non-oxidative electron transfer mechanism due to the interactions of GO-metal substrate and the contact interaction with the wrinkled GO surfaces, developing the oxidative stress efficiently complements the primary antibacterial activity. The generation of ROS (i.e. H_2O_2) by GO may cause significant

structural damage to the cell membranes by disrupting respiration, energy transduction, energetic balance of phospholipids, and transport of biologically active substances [84] (Fig. 12a).

In general, native PVP exhibits no antibacterial activity [85]. Due to the high surface energy (63.4 mJ.m^{-2}) and high surface tension (42.0 mJ.m^{-2}) of PVP, a mono-molecular layer of water is spontaneously formed over GP specimens while exposed to bacterial culture. As the hydrophilic surface affinity with water is stronger than that of the organic molecules, no conditioning layer is formed (Fig. 12b). Chen et al. [86] also demonstrated that high surface hydration leads to low fouling. When a hydrophilic polymer/surface comes in contact with water molecules, networks of hydrogen-bonds are formed inside the film due to the penetration of water molecules. With long-term exposure at the interface of the coated surface, the polymer swells because of hydrogen bonding with water molecules. The presence of ionic salts in bacterial culture can enhance the kinetics of swelling. So with such a hydrophilic surface, the interface is defined by the top water layer which inhibits the formation of a stable conditioning layer and maturation of biofilm.

Arkatkar et al. [87] reported that *Pseudomonas* sp. and *Bacillus* sp. biofilms are relatively hydrophobic, produce bio-surfactant, and form biofilm with comparatively higher carbohydrate and protein content. The pyrrolidone ring of PVP, being neutral and hydrophilic, is mostly responsible for the low interaction between protein and carbohydrate molecules which explains the conservation of the low protein adsorbing properties of the polymer film [88]. Our results on protein adsorption also strongly support the fact that hydrophilic surfaces exhibit a lower

adhesion of biomolecules. Our findings on GP specimens are in agreement with the above reports [86,87], resulting in reduced bio-adhesion and bio-proliferation on GP specimens (Fig. 12c).

4. Conclusions

An industrially viable and cost-effective GO/PVP composite coating on 316L SS, with excellent antibacterial and anti-biofouling activity, was developed using the electrophoretic deposition (EPD) technique. At an optimal EPD condition of 10 V, 60 s, and 0.03 wt% PVP, a uniform coating is achieved. The absence of —COOH peaks in GO and GP specimen, in XPS and FTIR data, confirm the partial reduction of GO due to the oxidative decarboxylation during the EPD process. The shift in the >N—C=O bond to the higher binding energy as compared to the pure PVP indicated the presence of hydrogen bonding between PVP and GO. The shift of Raman peak by $6\text{--}10\text{ cm}^{-1}$ in GP specimen indicated a higher distortion of GO structure on grafting with PVP. The composite coating showed an enhanced antibacterial activity by four orders for both *Pseudomonas* sp. and *Bacillus* sp. The excellent antibacterial activity and anti-biofouling properties of the new composite coating are attributed to the contact killing efficiency of GO, ROS mediated oxidative stress, and poor bio-adhesion on PVP surface due to surface hydration. The results show promising antibacterial and anti-biofouling applications of GO/PVP composite coating in seawater environments.

CRedit authorship contribution statement

Geetisubhra Jena: Conceptualization, Methodology, Software, Visualization, Investigation, Writing - Original draft preparation.

S. Sofia: Investigation.

B. Anandkumar: Supervision, Writing - Reviewing and Editing.

S. C. Vanithakumari: Supervision, Writing - Reviewing and Editing.

R. P. George: Supervision, Writing- Reviewing and Editing.

John Philip: Supervision, Writing - Reviewing and Editing.

Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2021.106356>.

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